Group Project

G4

Introduction

This project focuses on applying data analysis techniques within the supply chain domain to support better decision-making and improve operational efficiency.

Therefore, this project aims to explore how data analysis tools and techniques can be used to extract meaningful insights, identify patterns, and highlight inefficiencies in supply chain processes

Project overview

The proposed project aims to analyze supply chain data using a combination of analytical tools, including Excel, SQL, Python, Tableau, and Power BI.

The project will involve data cleaning, transformation, and analysis to evaluate key aspects.

Additionally, interactive dashboards will be developed to visualize KPIs and provide clear insights that can support decision-making.

The expected outcome of this project is to identify trends, detect potential, and provide actionable insights to enhance overall supply chain performance.

Objectives

Objectives

1

To analyze supply chain data in order to identify patterns, trends, and key factors affecting sales, inventory, and delivery performance.

2

To evaluate operational efficiency by detecting issues such as delays in shipments, stock shortages, and overstock situations.

3

To develop interactive dashboards and visualizations using tools like Power BI and Tableau to support data-driven decision-making.

The team

team leader: Manar Ashraf

The team

Abdelrhman mohamed hussien

Power BI

Manar Ashraf ESSA

Python/ Tableau

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Power BI

Abdelrahman Salah Abdelkareem

Excel/ SQL

Eman Salah Hassan

Excel/ Tableau

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Excel/ SQL

timeline

Timeline

The project will be completed over a period of 5 weeks, following a structured plan to ensure efficient progress:

Week 1: Data collection and data cleaning using Excel and SQL (Done).

Week 2: Data exploration and analysis using SQL and Python to identify key patterns and trends (in process).

Week 3&4: Data visualization and dashboard development using Power BI, Tableau, and Excel.

Week 5: Interpretation of results, generating insights, and final documentation.

Process

1. Data Collection:

Gathering supply chain data from available datasets, including orders,returns, and shipping data.

2. Data Cleaning & Preparation:

Handling missing values, removing duplicates, and transforming data into a usable format.

3. Data Analysis:

Using SQL and Python to explore the data, identify trends, and perform analytical operations.

4. Data Visualization:

Creating interactive dashboards using visualization tools to present key insights clearly.

5. Insights & Recommendations:

Interpreting the results and providing actionable recommendations to improve supply chain performance.

Literature Review

Supply chain

Supply chain analytics has become a critical area in modern business operations, as organizations increasingly rely on data to improve efficiency and reduce costs. It involves analyzing data generated from various supply chain activities such as procurement, inventory management, transportation, and customer demand.

Previous studies have shown that effective data analysis can help organizations identify inefficiencies, reduce delivery delays, and optimize inventory levels. By leveraging analytical tools, companies can make informed decisions that enhance overall performance and customer satisfaction.

Additionally, the use of visualization tools such as dashboards has proven to be highly effective in presenting insights in a clear and actionable way. These tools enable decision-makers to monitor KPIs and respond quickly to changes in supply chain operations.

Requirements Gathering

functional requirements

- A supply chain datasets, including orders, products, customers, and shipments.
- A set of data cleaning and transformation tools to ensure data quality.
- analyze key metrics .
- generate interactive dashboards and visual reports using visualizing tools.

System

system components

- Data Sources: Supply chain datasets including orders, products, customers, and shipping information.
- Data Processing Layer: Data cleaning and transformation using Excel and SQL.
- Data Storage & Querying: SQL is used to store and query structured data efficiently.
- Data Analysis Layer: Python is used for deeper analysis and pattern identification.
- Visualization Layer: Power BI and Tableau are used to create interactive dashboards and reports.

system Design

- Data collection from raw datasets.
- Data cleaning and preprocessing.
- Data storage and querying using SQL.
- Data analysis using Python.
- Data visualization and reporting.

THANK YOU